AgriSense

कृषि मूल्य पूर्वानुमान प्रणाली | Crop Price Intelligence

AI-Powered Price Forecasting Across 34 Indian States & 22 Commodities

507,263
Data Points

0.49%
Avg Model Error

99.51%
Accuracy

7-Day
Forecast Horizon

🌐 Live Dashboard

Ravi Shekhar Sharma · C&A Winter Project · 2025

SITUATION

India's Crop Markets Are Volatile, Fragmented & Data-Blind

600M+

Farmers dependent on crop markets for income

49.6%

Tomato price swings (Coefficient of Variation)

40%

Post-harvest losses due to poor price information

Without accurate, real-time price forecasting, farmers sell at wrong times, traders exploit information asymmetry, and policymakers react to crises rather than preventing them. The result: prices that bear little relation to supply and demand fundamentals.

The gap: Market data exists. Forecasting models exist. A unified, accessible intelligence layer for India's agricultural ecosystem does not.

DATA FOUNDATION

A National Dataset: 507,263 Daily Price Observations

507,263
Total Observations
Daily prices, Aug 2022–Aug 2024

34
Indian States & UTs
Full national coverage

22
Commodities Tracked
Cereals, pulses, oils, vegetables

732
Days of Data
2 full years, daily frequency

COMMODITY COVERAGE

CEREALS
Rice · Wheat · Atta

PULSES
Tur Dal · Urad Dal · Moong Dal
Gram Dal · Masoor Dal

VEGETABLES
Tomato · Onion · Potato

EDIBLE OILS
Mustard · Groundnut · Soya
Sunflower · Palm · Vanaspati

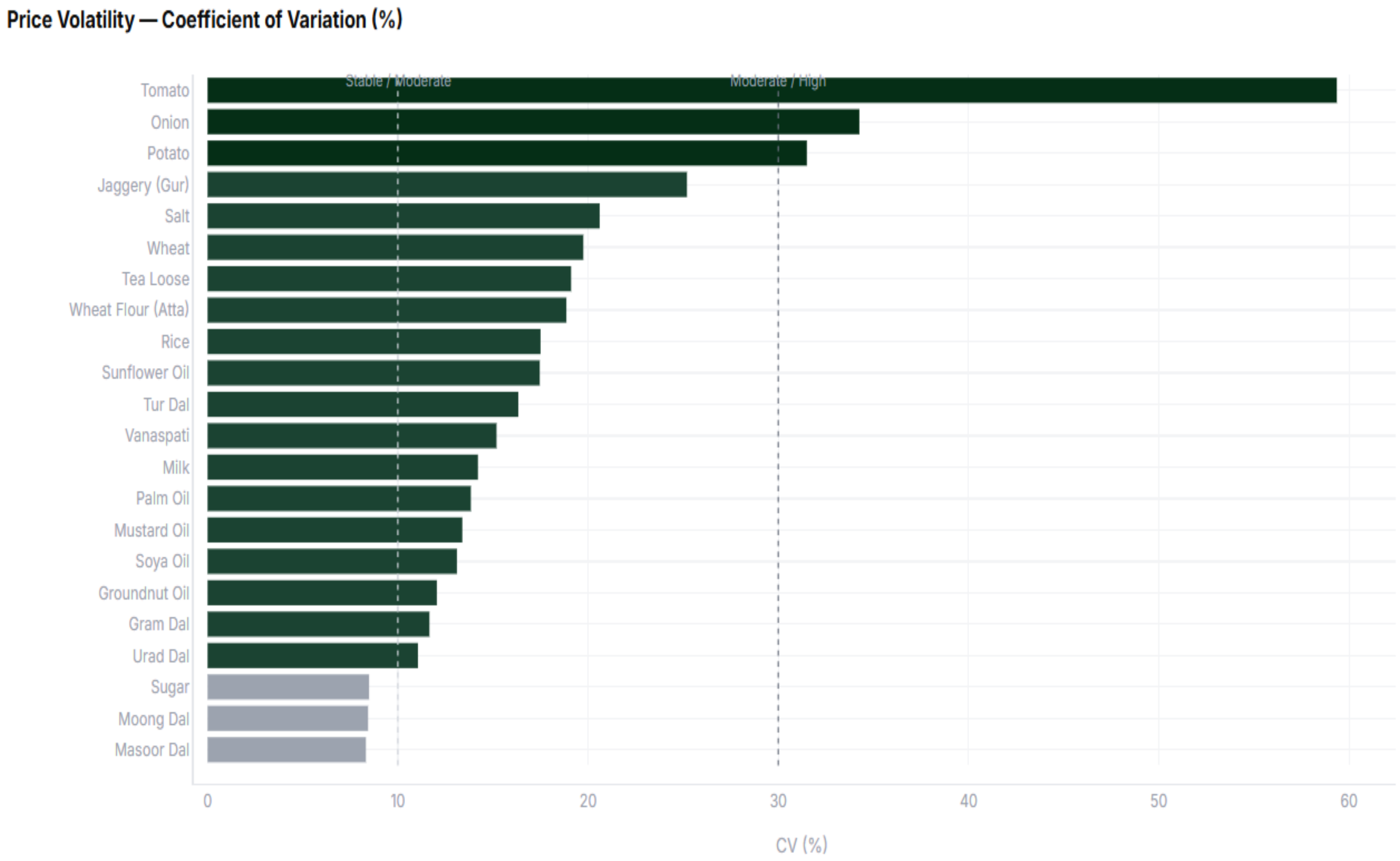
SUGAR & DAIRY
Sugar · Gur · Milk

OTHERS
Tea (Loose) · Salt

MARKET ANALYSIS

Commodity Volatility: Tomato Is 17× More Volatile Than Salt

PRICE VOLATILITY — COEFFICIENT OF VARIATION (%)



KEY INSIGHTS

Tomato (49.6% CV)

Most volatile — driven by dual crop seasons, perishability and monsoon disruptions.

Onion (26.1% CV)

Structurally volatile — Kharif vs Rabi supply cycles create predictable spike windows.

Potato (17.0% CV)

Moderate — August peak aligns with storage depletion before new crop arrival.

Salt & Sugar (<5%)

Government-regulated with buffer stocks — near-zero volatility.

TASK

AgriSense: A Three-Layer Market Intelligence System

01 Predict

Machine learning models forecast retail prices up to 7 days ahead, per commodity per state — trained on 2 years of daily price history.

- **22 Commodities**
- **34 States**
- **7-Day Horizon**

02 Analyse

Interactive dashboard surfaces price trends, seasonal patterns, festival demand signals, and regional price dispersion across India.

- **6 Dashboard Tabs**
- **Real-Time Filters**
- **Live URL**

03 Act

Margin anomaly detection flags states with abnormal retail-wholesale spreads — identifying supply chain disruptions and arbitrage opportunities.

- **1.5σ Anomaly Threshold**
- **State-Level Alerts**
- **Downloadable Reports**

METHODOLOGY

End-to-End ML Pipeline: From Raw Data to Live Forecast

1
Data Ingestion
Retail + Wholesale
CSVs (raw)

2
Preprocessing
Cleaning · Normalisation
Missing value imputation

3
Feature Engineering
84 features across
9 categories

4
Model Training
22 XGBoost models
(per commodity)

5
Evaluation
RMSE · MAE · MAPE
R^2 on test set

6
Dashboard
Streamlit · Live URL
6-tab interface

CRITICAL DESIGN DECISIONS

Time-Based Split
Train Aug'22–Jan'24 · Test Feb–Aug'24. No random split — prevents data leakage through lag features.

Per-Commodity Models
22 independent models. Tomato (CV 49.6%) and Salt (CV 2.9%) cannot share model parameters.

Impute Before Lag
Missing values filled (ffill → bfill → interpolation) before computing lags. Prevents NaN propagation.

84 Features Engineered Across 9 Analytical Dimensions

8	**Lag Features**	1 · 7 · 14 · 30-day retail & wholesale lags
15	**Rolling Statistics**	7d & 30d mean, std, z-score, min/max range
4	**Momentum**	7d & 30d % price change (trend direction)
18	**Calendar**	Year, month, day, cyclical sin/cos encoding
5	**Spread & Margin**	Retail–wholesale spread, ratio, 7d/30d rolling
6	**Harvest Flags**	Is-harvest-month per crop, Kharif/Rabi seasons
11	**Festival Calendar**	Diwali, Holi, Eid, Navratri, Pongal proximity
7	**Category Encoding**	One-hot: Cereal/Pulse/Vegetable/Oil/Sugar/Dairy
7	**Regional Encoding**	North/South/East/West/Central/Northeast regions

Total: 84 features · Input: merged_retail_wholesale.csv (wide format, 23,374 rows) · Output: features.csv (507,263 rows)

WHY THESE FEATURES

Lag features
Prices are autocorrelated — yesterday's price is the single strongest predictor of today's.

Cyclical encoding
Dec and Jan are neighbours. Raw month=12 and month=1 look 11 apart to a model without sin/cos.

Festival calendar
Diwali spikes oil/sugar demand. Navratri drops onion demand. These are structural, predictable events.

Harvest flags
Supply peaks → price falls. Knowing a commodity is in harvest month helps predict price direction.

Margin features
Retail–wholesale spread captures supply chain friction independent of absolute price level.

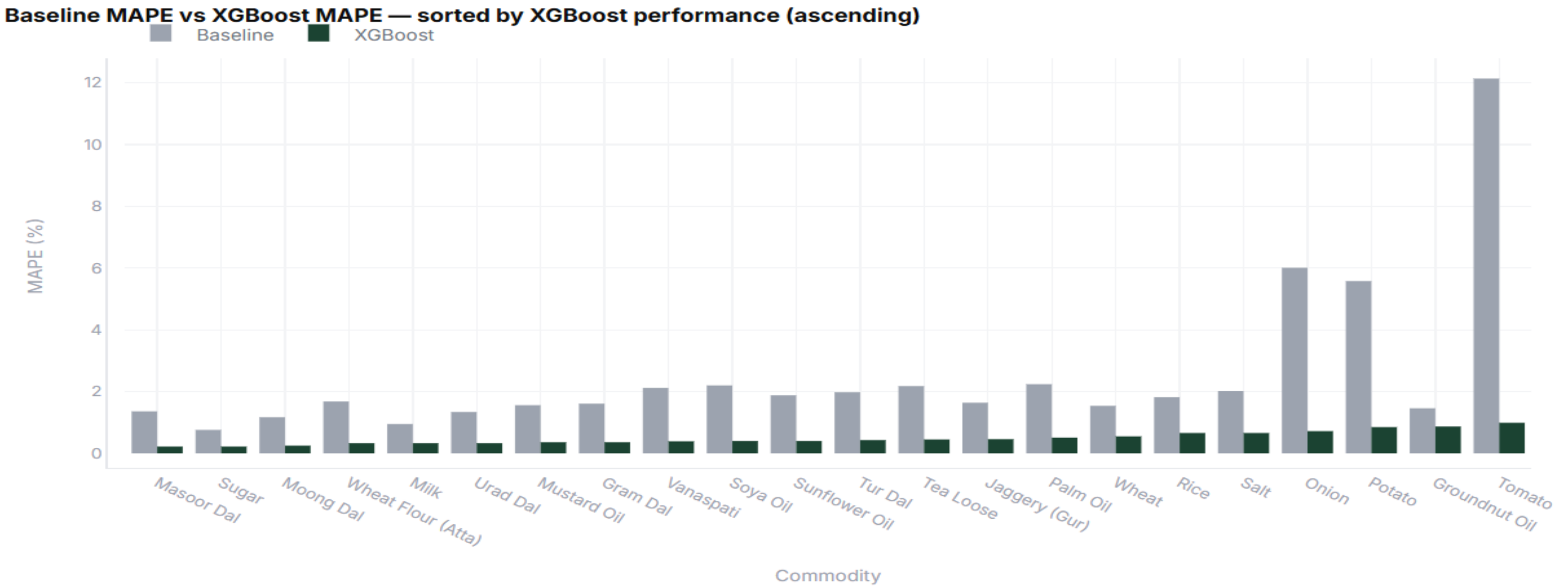
XGBoost Reduces Forecasting Error by 80% Over Naive Baseline

0.49%

Average MAPE

Mean Absolute % Error across all 22 commodities on held-out test set

MAPE COMPARISON — BASELINE VS XGBOOST (ALL 22 COMMODITIES)



0.994
Average R^2
Variance explained

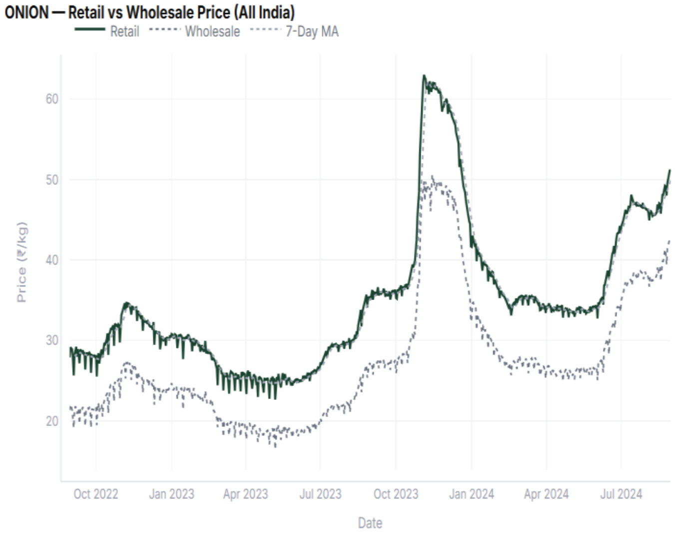
66
Models Trained
22 commodities × 3 types

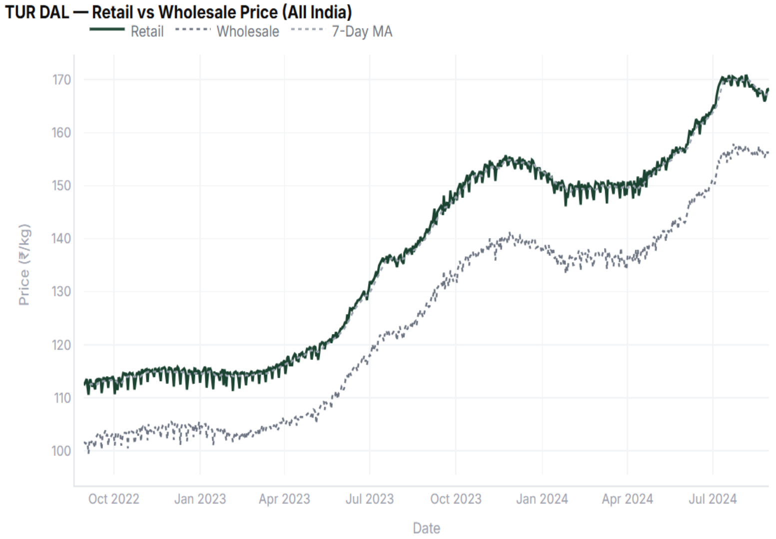
1.1
Minutes to Train
All 66 models

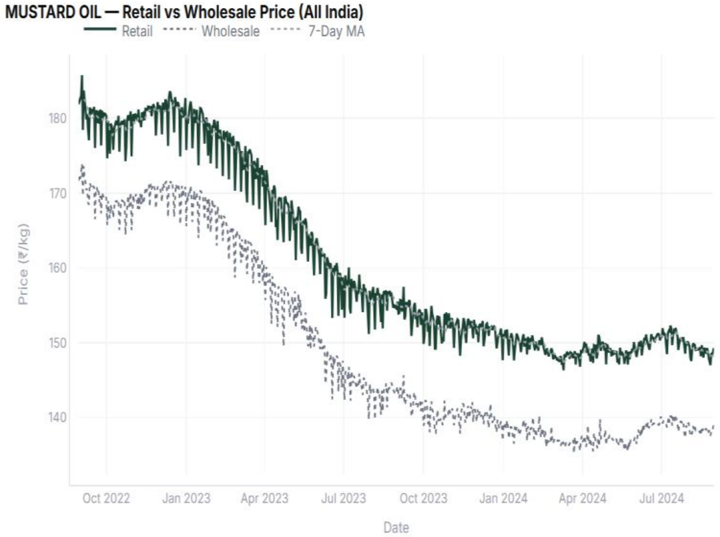
5×
Better than Baseline
2.51% → 0.49% MAPE

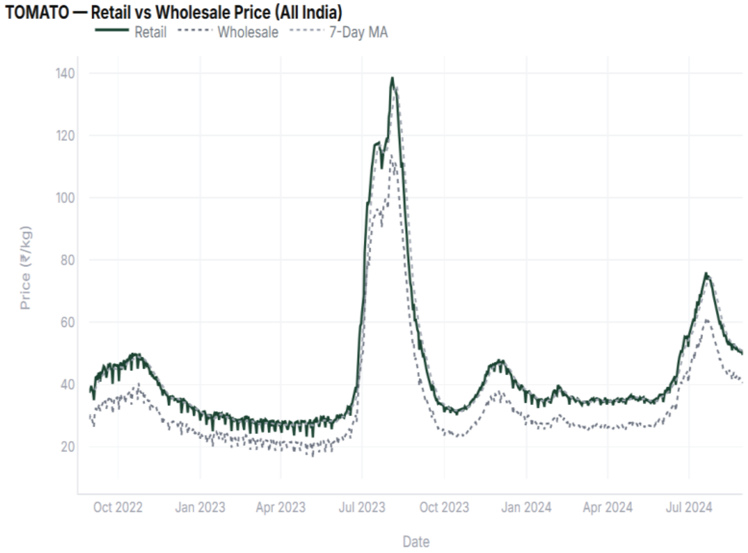
MARKET INTELLIGENCE

Price Movements Reveal Structural Patterns Across India









STRUCTURAL FINDINGS

+78% **Onion**

Kharif crop failure + export ban in 2023 created the sharpest price spike in the dataset.

+49% **Tur Dal**

Consecutive drought years in Maharashtra + pulse import restrictions drove sustained price rise.

–18% **Mustard Oil**

Only commodity to decline — driven by record Rabi crop output and palm oil import policy shift.

July **Tomato Peak**

Monsoon disrupts supply chain every July — the most predictable seasonal pattern in the dataset.

PRODUCT

Live Dashboard: Intelligence for Three Stakeholder Groups

Farmers
किसान

Dashboard: 7-Day Forecast

See predicted retail prices for their crop 7 days ahead — enabling optimal harvest timing and mandi selection.

- Forecast Horizon: 7 days
- 22 Crops Covered
- State-Level Granularity

Traders & Wholesalers
व्यापारी

Dashboard: Margin Opportunity

Identify states with abnormally high retail-wholesale margins — signalling arbitrage opportunities or supply chain disruptions.

- 1.5σ Anomaly Detection
- All 34 States Ranked
- Downloadable Reports

Policymakers
नीति निर्माता

Dashboard: State Heatmap

Real-time view of price dispersion across India — early warning for where government buffer stock interventions are needed.

- National Price Map
- Regional Analysis
- Historical Trends

🌐 **Live Dashboard:** agrisense-crop-price-intelligence.streamlit.app

TECHNICAL STACK

Production-Grade Architecture Built for Scale and Modularity

Data & Processing	Machine Learning	Dashboard & Deployment	Version Control
Python 3.12 Core language	**XGBoost** Primary model — gradient boosted trees	**Streamlit** Interactive web dashboard framework	**Git + GitHub** Source control and repo hosting
Pandas + NumPy Data manipulation & numerical ops	**LightGBM** Benchmark — fast gradient boosting	**Plotly** Interactive data visualizations	**requirements.txt** Reproducible environment spec
Scikit-learn Label encoding, train/test split	**Early Stopping** Validation-based overfitting prevention	**Streamlit Cloud** Live hosting, free tier	**Modular src/** Preprocessing, FE, training as modules

Modular Design: Add a new commodity = 1 line in config. Swap model = change 1 variable. All logic flows through top-level config dicts — zero hardcoding inside functions. Production-ready architecture.

ROADMAP

What AgriSense Becomes With More Data and Resources

Phase 1

Current State

- 2 years historical data
- 22 commodities, 34 states
- XGBoost per-commodity model
- 7-day autoregressive forecast
- Streamlit dashboard, live URL

Phase 2

Near-Term

- Daily model retraining pipeline
- Rainfall & weather feature (IMD API)
- MSP policy announcement signals
- WhatsApp/SMS price alert integration
- Mobile-responsive dashboard

Phase 3

Production Scale

- All 550+ APMC mandi markets
- Real-time price feeds (Agmarknet API)
- Hierarchical Bayesian model
- District-level granularity
- API for third-party integrations

AgriSense (एग्रीसेंस)

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0.49%	**0.994**	**66**	**Live**
Avg MAPE	Avg R^2	Models	Deployed

Live Dashboard agrisense-crop-price-intelligence.streamlit.app

GitHub Repository github.com/Ravi-Shekhar-Sharma/agrisense-crop-price-intelligence